

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Have you worked on client-side programming before?



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can mention the two aspects ASP.NET Client side coding has.
- I can talk about Client side scripts.
- I can discuss ideas about Client side source code.

Ask your partner:

- What is the difference between server-side and client-side programming?
- Have you ever coded a server-side tool before? How?





Let's Explore!

We need to work further on Client Side Programming in ASP.NET.



Instructions: Read the text below.

Lecturer: So far, we have been exploring different ASP.NET programming features, but we haven't given much attention to client-side programming on this platform. You must know that ASP.NET pages and applications are not restricted to server-based tools or languages; there is also a wide variety of client-based tools available in ASP.NET. So let's start with a simple question: what aspects are included in client-side coding?

Well, there are essentially two aspects that are pretty important in this type of programming: client-side scripts, on the one hand, and side source coding, on the other. Client-side scripting makes reference to those programs that are run on the web by the user's web browser instead of using the web server; this type of scripts allows speeding up the page execution. So, ASP.NET provides different server controls that use client-side scripting in order to respond to the users' request without needing to send a post back to the server. One clear example of this type of processing can be client-side data validation; for instance, a user can fill and validate the information in an email form field without making a round trip to the server.

ASP.NET developers have been working hard to make server controls be able to execute scripts through different events; for instance, onblur, onfocus, onchange, onkeydown, onkeyup, and onserverclick, among others.

But, how is client source code written? Well, as we have discussed in recent sessions, there are specifically two types of files in which ASP.NET pages are written: the content file or markup file (.aspx) and the code-behind file. The content file holds the structure of a page, which is the whole of control tags of the page; and the code-behind file has the class definition. Once both files are executed, the content file is transformed into a page class; then, as a result of the combined data of this class, the class definition, and the system generated code, we get the executable code or assembly to process clients' requests.



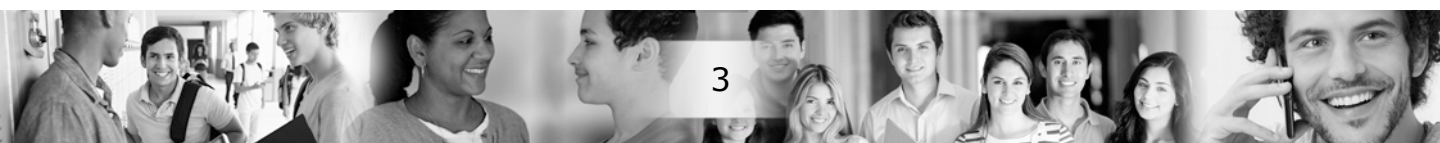
Let's Practice!

These are some aspects about client-side scripts in ASP.NET.

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. We have learned that client-side scripting programs run on the web by the user's web browser instead of using the web server.
2. ASP.NET developers have been working hard to make server controls be able to execute scripts through different events.
3. We have been studying client-side scripts and side source coding in our programming class.
4. The lecturer has explained to us that ASP.NET provides different server controls that use client-side scripting.
5. She has mentioned different examples to explain client-side scripting, for instance, client-side data validation.
6. The lecturer has been giving us different bibliographic sources to identify how to code events such as onblur, onfocus, onchange, onkeydown, onkeyup, and onserverclick, among others.

Present Perfect Simple	Present Perfect Continuous





Let's Practice!

How much have you learned about client-side source code?

Instructions: Select the correct answer.

1. As we _____ in class, there are two types of files in which ASP.NET pages are written: the content file or markup file (.aspx) and the code-behind file.
 - a. discusses
 - b. has discussed
 - c. has been discussing
 - d. have been discussing
2. Is the structure of the page _____ on the content file?
 - a. hold
 - b. held
 - c. holds
 - d. holding
3. The system _____ the content file into a page class.
 - a. transformed
 - b. has transformed
 - c. have transformed
 - d. have been transforming
4. How is the executable code or assembly _____?
 - a. generate
 - b. generates
 - c. generated
 - d. generating
5. The combined data of these files _____ the assembly.
 - a. creates
 - b. created
 - c. has created
 - d. have created



Extension Activity

Getting Acquainted with Client Side Coding in ASP.NET!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about client-side coding in ASP.NET.
- Summarize the main aspects about client-side scripts and client side-source code.
- Prepare a poster or a slide show presentation with this information.
- Explain the main aspects about client-side coding in ASP.NET to your class.

Self-Assessment

- I search for more information about client-side coding in ASP.NET.
- I summarize the information.
- I organize the information in a poster or slide show presentation.
- I explain the main aspects about client-side coding to my class.

Yes

No

☐
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Target Words

- class definition** (noun): the setting of methods and variables that will be used in an object
e.g. The content file holds the structure of a page, which is the whole of control tags of the page; and the code-behind file has the class definition.
- assembly** (noun): a runtime or deployment unit
e.g. As a result of the combined data of this class, the class definition, and the system generated code, we get the executable code or assembly to process clients' requests.
- script** (noun): a series of instructions or commands that need to be interpreted by a specific program
e.g. ASP.NET developers have been working hard to make server controls be able to execute scripts through different events.

- mark-up file** (noun): a file that possesses all the logical structure of pages
e.g. There are specifically two types of files in which ASP.NET pages are written: the content file or markup file (.aspx) and the code-behind file.
- post back** (noun): the process of sending an ASP.NET page to the server for processing
e.g. So, ASP.NET provides different server controls that use client-side scripting in order to respond to the users' request without needing to send a post back to the server.



Learning Points

Present Perfect Simple vs. Present Perfect Continuous in Affirmative Form

You use the **present perfect** and the **present perfect continuous** tenses to talk about actions that started in the past but are still in progress or have just finished in the present. However, there is a remarkable difference between the meanings of both tenses.

When you use the **present perfect simple**, you want to indicate that an action is completed or aim to highlight the result of an action.

Examples:

- She **has** already **finished** her presentation about client-side coding. (completed action)
- That programmer **has used** client-side scripts to code the controls on that page. (what)
- My students **have sent** me thousands of questions about client-side programming. (how much)
- I **have read** that article about client-side programming twice. (how often)

On the other hand, you use the **present perfect continuous** to highlight the duration of an action that is in progress.

Examples:

- The programmer **has been working** on that code for hours. (how long)
- **I have been preparing** a presentation about client-side coding. (That's what I've been doing, it doesn't matter if the presentation is finished or not)

The following are the structures you need to use when building affirmative sentences with both structures:

Present Perfect Simple

Subject + **have/has** + **main verb in past participle** + complement.

Present Perfect Continuous

Subject + **have/has** + **been** + **main verb in ING** + complement.

It's important to note that there are some verbs that cannot be used with the **present perfect continuous** but only with the **present simple tense**. Take a look at the chart below:





Learning Points

Thinking	Senses	State
• Think • Believe • Know • Understand	• Feel • Smell • Taste • See • Hear • Touch	• Be • Have (possession)

Examples

- *So far, we have been exploring different ASP.NET programming features, but we haven't given much attention to client-side programming on this platform.*
- *ASP.NET developers have been working hard to make server controls be able to execute scripts through different events.*
- *Well, as we have discussed in recent sessions, there are specifically two types of files in which ASP.NET pages are written: the content file or markup file (.aspx) and the code-behind file.*

Simple Present Passive in Questions

We use sentences in **simple present passive** or **present passive voice** when we want to stand out the function of an **object** in a sentence rather than the **subject**. This tense is commonly used when we expect to provide explanations.

When we aim to highlight the person or element that performs an action, we use active voice.

Examples:

- **ASP.NET pages and applications** use client-based tools.
- *Do ASP.NET pages and applications use client-based tools?*

But when it comes to highlighting the role of the object in a sentence, we need to use the *passive voice*.





Learning Points

Examples:

- **Client-based tools** are used by ASP.NET pages and applications.
- Are **client-based tools** used by ASP.NET pages and applications?

When using the passive voice the subject could be omitted, but in case you wish to mention it, you need to use the particle **by** followed by the noun.

This is the grammatical structure we need to use when building **interrogative** sentences in **simple present passive**:

Wh-word (optional) + **Verb be** + Object +
Verb in past participle + Complement?

Examples:

- **Are** the client-side scripts **executed** in the browser or in the server?
- **Is** client-side coding **provided** by ASP.NET pages?
- Where **are** the client-side scripts **executed**?
- How **are** events **raised** by server controls?

Examples

- What aspects are included in client-side coding?
- How is client source code written?